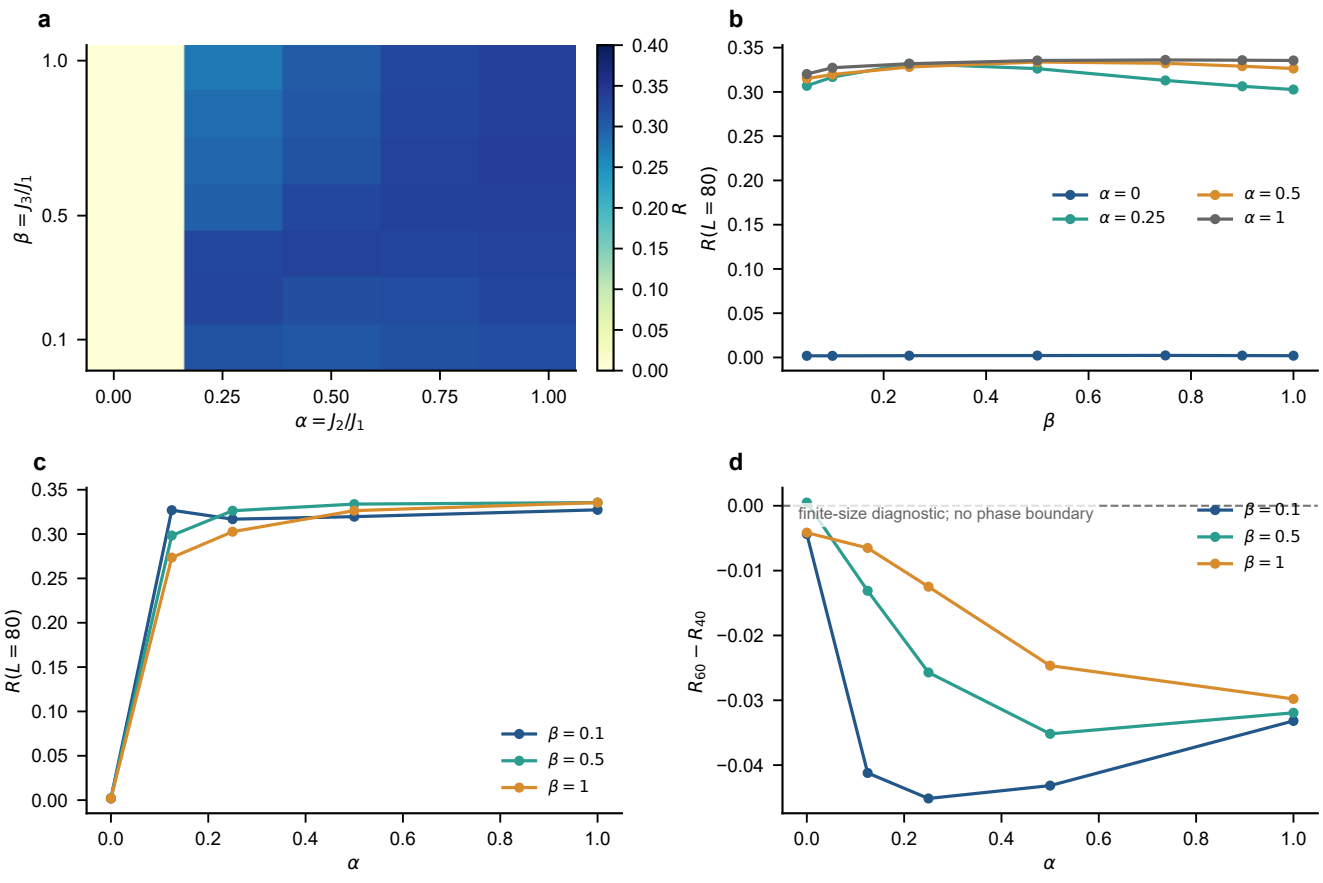
a
b
c
d
$\beta = J_3/J_1$
$\alpha = J_2/J_1$
R
$R(L = 80)$
β
α
$R_{60} - R_{40}$
$\alpha = 0$
$\alpha = 0.25$
$\alpha = 0.5$
$\alpha = 1$
$\beta = 0.1$
$\beta = 0.5$
$\beta = 1$
finite-size diagnostic; no phase boundary